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GPU DESIGN USING SIMD ARCHITECTURE

Introduction

Graphics Processing Units (GPUs) are specialized hardware components designed to handle **parallel processing tasks** efficiently. Unlike Central Processing Units (CPUs), which are optimized for sequential execution, GPUs can execute thousands of operations simultaneously, making them highly effective for computationally intensive tasks such as graphics rendering, machine learning, scientific simulations, and cryptographic applications.

One of the fundamental architectures used in GPUs is **SIMD (Single Instruction, Multiple Data)**. SIMD enables the execution of a single instruction on multiple data elements in parallel. This significantly improves throughput, reduces computation time, and enhances performance in tasks where the same operation must be applied to large datasets.

Area of Application / Work / Domain

I. **Domain:** Computer Architecture and Parallel Processing

II. Applications:

A. Graphics rendering (games, simulations)

B. Digital Image Processing

C. Machine Learning and AI

D. Scientific Computing

This project demonstrates a **basic SIMD-based GPU concept** using a block diagram and flow representation. The focus is on understanding how SIMD improves performance in tasks like vector operations, which are common in image and signal processing.

Literature Survey

- **Flynn's Taxonomy (1966):** Classified processors into SISD, SIMD, MISD, and MIMD. GPUs are mainly SIMD/MIMD.
- **Early Vector Processors:** Machines like Cray supercomputers applied SIMD principles.
- **Modern GPUs:**
  - NVIDIA CUDA programming model (2006) expanded GPU use beyond graphics.
  - AMD GCN and RDNA architectures rely heavily on SIMD compute units.
- **Research Findings:** SIMD improves speed and efficiency for repetitive tasks like image filtering, signal processing, and neural network computations.
- Flynn's taxonomy provides a theoretical base to understand parallel processing.
- Early vector processors proved that SIMD can boost performance for repetitive tasks.
- Modern GPUs (NVIDIA, AMD) use SIMD to handle data-parallel workloads efficiently.
- Research shows SIMD-based GPU computing is faster and more energy-efficient than sequential CPU processing for large datasets.

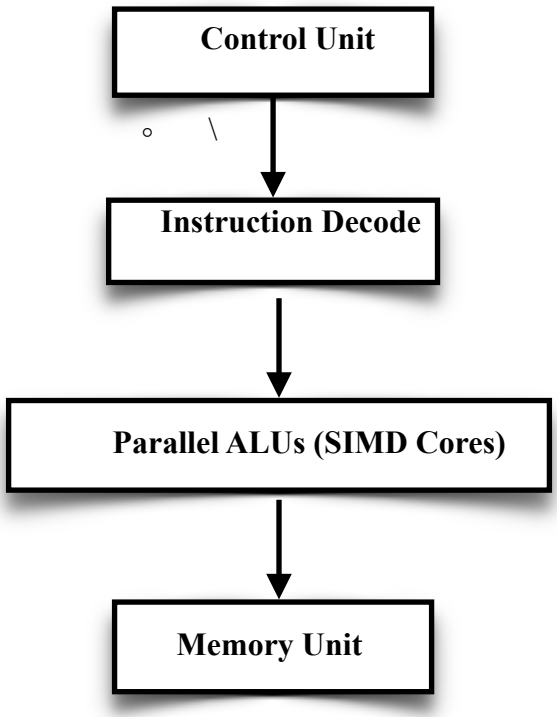
- Flynn's taxonomy provides a theoretical base to understand parallel processing.
- Early vector processors proved that SIMD can boost performance for repetitive tasks.

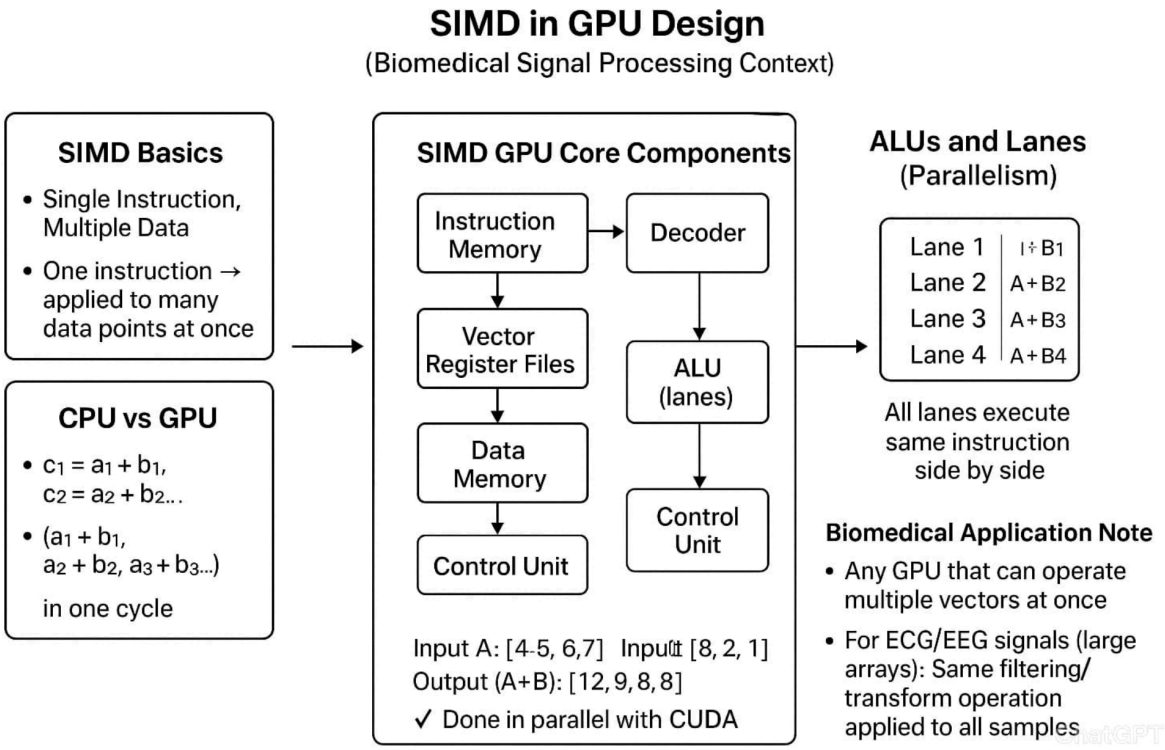
Problem Definition

- CPUs are efficient for **sequential tasks** but not for highly parallel workloads.
- Applications like image processing and machine learning require handling **large datasets quickly**.
- Challenge: To design a simple architecture that shows how GPUs perform better than CPUs using SIMD.

Proposed Solution / Block Diagram

- Design a **basic SIMD GPU model** showing:
  - **Control Unit** → issues instruction
  - **Instruction Decoder** → converts instruction
  - **Parallel ALUs (SIMD Cores)** → perform operation on multiple data
  - **Memory Unit** → for data storage and retrieval





Project Execution Steps

1. Study of GPUs and SIMD basics.
2. Design of block diagram showing SIMD architecture.
3. Selection of a sample task (Vector Addition).
4. Create a simple flowchart to explain SIMD operation.
5. (Optional) Simulation using Verilog / C / MATLAB for vector addition.
6. Prepare documentation and presentation.

Estimated Bill of Materials (if implemented on hardware)

- Software Tools (Free): Verilog, MATLAB (Student version).
- Hardware (Optional): FPGA Development Board (use college lab equipments).
- Documentation (Printing, Binding)

  Total Cost (Software-only version): Minimal (~₹500).

Hardware-Oriented Timeline (FPGA-based Implementation)

Week	Tasks	Expected Output
Week 1	Study of GPU basics, SIMD concept, and FPGA fundamentals	Theoretical understanding & reference notes
Week 2	Write simple **Verilog/VHDL code** for SIMD unit (e.g., 4 parallel ALUs for addition)	Draft HDL code
Week 3	Test HDL code in simulation software (ModelSim/Quartus)	Simulation waveforms of vector addition
Week 4	Synthesize design and map to FPGA board (if available)	Bitstream file generated
Week 5	Hardware testing on FPGA (input/output verification using switches/LEDs or UART)	Working hardware prototype
Week 6	Documentation + presentation	Final report + demonstration

IEEE References

- **M. J. Flynn**, "Some Computer Organizations and Their Effectiveness," *IEEE Transactions on Computers*, vol. C-21, no. 9, pp. 948–960, Sept. 1972. ResearchGate

- **J. L. Hennessy** and **D. A. Patterson**, *Computer Architecture: A Quantitative Approach*, 5th ed., Morgan Kaufmann, 2011. acm-prod.literatumonline.comElsevier Educate

- **NVIDIA Corporation**, *CUDA C Programming Guide*, Version 12.0, NVIDIA Developer, 2023.
  (Since this is a technical manual rather than a peer-reviewed article, you can cite it as above.)

- **AMD**, *RDNA™ Architecture Whitepaper*, Advanced Micro Devices, 2019.
  (Similarly, treat this as a credible industry source.)

Why Flynn (1972)?

This foundational paper introduces **Flynn's taxonomy** (SISD, SIMD, MISD, MIMD), helping explain why SIMD is efficient for parallel tasks—your project's core idea. ResearchGate

Why Hennessy & Patterson (2011)?

Their book is a trusted reference in computer architecture. It covers SIMD, vector processing, GPU design, and performance analysis—lending credible academic weight to your project. acm-prod.literatumonline.com

Why include NVIDIA and AMD sources?

These provide real-world examples of SIMD architecture in action (CUDA for NVIDIA GPUs, RDNA for AMD). Even though they're whitepapers or guides, they add practical relevance and show you're familiar with current industry trends.

Guide recommendation

Remarks from reviewers